The Triadic Redox Theory: A Unified Framework for Electron Transfer

Macharia Barii

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Abstract

Every chemistry, every biological reaction, and every energy-storage device is, at its root, the same phenomenon: the movement of electrons from a donor to an acceptor through a mediating medium. Despite this centrality, the disciplines that study electron transfer—electrochemistry, corrosion science, catalysis, and biochemistry—remain separated by distinct theories, vocabularies, and methods. Here we present the **Triadic Redox Theory (TRT)**, a unified framework that classifies every electron-transfer interaction through three archetypal states: the **Hydrogen-like (H)** electron donor, the **Helium-like (He)** electron acceptor, and the **Middle Fluid (Φ)** that mediates, modulates, or blocks the transfer. We define three quantitative state functions—χ_{H} (donor tendency), χ_{He} (acceptor tendency), and Φ (Fluidity, the field-derivative of the dielectric response)—and derive a single fundamental inequality governing all spontaneous electron transfer:

$$\chi_{\mathrm{H}}(\text{Donor}) < \Phi(\text{Medium}) < \chi_{\mathrm{He}}(\text{Acceptor}). \tag{1}$$

From this inequality we derive the **Operational Window** ($0.5 < \Phi < 2.0$) within which electron-transfer systems are stable and functional, and a hierarchy of application-specific windows—the Fluidity Window (batteries), the Corrosion Potential Window (corrosion), the Catalytic Activity Window (catalysis), and the Biological Fluidity Window (life). We develop the kinetic formalism connecting Fluidity to reorganization energy and establish the **Law of Fluid Resonance** ($\lambda \approx \eta$). We compile a state-function database for elements, solvents, and engineering materials, present a fully specified density-functional-theory (DFT) computational protocol, validate the inequality against known battery materials, and propose a three-dimensional property space ($\chi_{\mathrm{H}}, \chi_{\mathrm{He}}, \Phi$) for prescriptive materials design. TRT reduces the design-search problem from empirical trial-and-error to a single checkable inequality, with companion works applying the framework to battery design, corrosion, catalysis, biology and evolution, respiration and astrobiology, nuclear physics, and the structure of existence itself.

Keywords: Triadic Redox Theory, electron transfer, Fluidity, fundamental inequality, state functions, Operational Window, reorganization energy, density functional theory, materials design, unified theory.

1 Introduction

1.1 The electron as the universal currency

Electrons are the universal currency of energy transfer. They move from regions of high potential to regions of low potential, driving every reaction in the universe—from the fusion of stars to the beating of a human heart. All chemistry is electron-driven; all biology is chemistry; and therefore all biology is, at its foundation, the guided flow of electrons. A battery discharges, a metal corrodes, a catalyst assembles a molecule, and an enzyme powers a cell because electrons are transferred from places that give them up easily to places that accept them eagerly.

This observation, once stated, appears obvious. Yet its consequences are profound and largely unexploited: if all electron-transfer phenomena share a single mechanistic core, then they should be describable by a *single* unified framework. The purpose of this paper is to present that framework.

1.2 The failure of the partitioned approach

The conventional paradigm treats the components of an electron-transfer system as independent. Electrochemists study electrodes; corrosion scientists study environments; catalytic chemists study surfaces and complexes; biochemists study proteins—each with dedicated theories, terminology, and computational tools [4, 5, 6]. Three of the most powerful existing frameworks illustrate the problem:

- **Marcus theory** [1] describes the *kinetics* of electron transfer through the reorganization energy λ and the electronic coupling, but it does not address the *thermodynamic stability* of the coupled donor—medium—acceptor system, nor does it couple the three components into a single predictive condition.
- The **hard/soft acid–base (HSAB) principle** [2] provides a qualitative classification of reactivity but yields no quantitative voltage windows or material-selection rules.
- **Band theory** [3] describes the electronic structure of a single material, but does not predict whether an anode, an electrolyte, and a cathode will be mutually compatible.

Each framework is correct within its domain; none unifies the domains. The result is that materials discovery in batteries, corrosion protection, and catalysis remains dominated by empirical trial-and-error, with combinatorial search spaces far exceeding what can be realistically explored.

1.3 The insight: three archetypes

The periodic table provides the clue. Its two extremes—hydrogen, with a single valence electron it readily shares or loses, and helium, with a closed shell that makes it electron-hungry—represent the ultimate electron *donor* and the ultimate electron *acceptor*. Every other species displays intermediate behavior: it can donate electrons, accept electrons, or—critically—*reorganize its electron density* to facilitate the transfer between a donor and an acceptor.

The Triadic Redox Theory is built on the proposal that every electron-transfer interaction can be classified through three archetypal states:

Definition 1 (Hydrogen-like (H))**.** *A state characterized by a tendency to* donate *electrons. The archetypal example is the hydrogen atom, which has one valence electron that it readily shares or loses. Macroscopic analogues include anodes, reducing agents, and electron-rich metal surfaces.*

Definition 2 (Helium-like (He))**.** *A state characterized by a tendency to* accept *electrons. The archetypal example is the helium atom, which has a full valence shell and is therefore electron-hungry. Macroscopic analogues include cathodes, oxidizing acids, and electron-poor species such as molecular oxygen.*

Definition 3 (Middle Fluid (Φ))**.** *A state characterized by the* capacity to reorganize electron density*. The archetypal examples include solvents, catalysts, protein scaffolds, and electrolyte matrices. The Middle Fluid is neither donor nor acceptor: it is the mediator that allows, modulates, or blocks the transfer between them.*

These three states recur at every scale—from the quantum electron to the electrochemical cell to the ecosystem. They are the grammar of the universe of electron transfer. The remainder of this paper formalizes this proposal into measurable state functions (Section 2), a fundamental inequality (Section 3), a kinetic formalism (Section 4), and a hierarchy of operating windows (Sections 5 and 6), together with a computational protocol (Section 7), a unified property space (Section 8), and a set of cross-domain design rules and predictions (Sections 9–10).

2 State Functions

TRT assigns three quantitative state functions to every material, every solvent, and every medium. Each is computable from first principles and measurable experimentally.

2.1 The H-Character, χ_{H}

Definition 4 (H-Character χ_{H})**.** *The* ***H-Character*** *of a material is the energy (in eV) required to remove one electron from its highest occupied state:*

$$\chi_{\mathrm{H}} = -\varepsilon_{\mathrm{HOMO}} \quad \textit{(molecules)} \quad \textit{or} \quad \chi_{\mathrm{H}} = -E_{\mathrm{VBM}} \quad \textit{(solids)}. \tag{2}$$

Physically, χ_{H} is the donor tendency: the energy cost of ionization. A low χ_{H} means the material gives up electrons readily ("more H-like"); a high χ_{H} means it holds its electrons tightly ("more He-like").

- *Range:* $\approx 0\,\mathrm{eV}$ (metallic, pure H-like) to $> 5\,\mathrm{eV}$ (insulating).
- *Engineering guideline:* a material intended as an electron *donor* (anode, reducing agent) should satisfy $\chi_{\mathrm{H}} < 3.0\,\mathrm{eV}$.

2.2 The He-Character, χ_{He}

Definition 5 (He-Character χ_{He})**.** *The* ***He-Character*** *of a material is the energy (in eV) released when one electron is added to its lowest unoccupied state:*

$$\chi_{\mathrm{He}} = -\varepsilon_{\mathrm{LUMO}} \quad \textit{(molecules)} \quad \textit{or} \quad \chi_{\mathrm{He}} = -E_{\mathrm{CBM}} \quad \textit{(solids)}. \tag{3}$$

Physically, χ_{He} is the acceptor tendency: the energy gain on reduction. A high χ_{He} means the material accepts electrons eagerly ("more He-like").

- *Range:* $\approx 0\,\mathrm{eV}$ (inert, pure He-like at threshold) to $> 5\,\mathrm{eV}$ (highly oxidizing).
- *Engineering guideline:* a material intended as an electron *acceptor* (cathode, oxidant) should satisfy $\chi_{\mathrm{He}} > 4.5\,\mathrm{eV}$.

2.3 The Fluidity, Φ

Definition 6 (Fluidity Φ)**.** *The* ***Fluidity*** *of a medium is a dimensionless measure of its electronic polarizability and reorganizational capacity, defined as the field-derivative of its relative dielectric constant:*

$$\Phi = \frac{1}{\varepsilon_0}\frac{\partial \varepsilon_r}{\partial E}, \tag{4}$$

where ε_r is the relative dielectric constant, E the applied electric field, and ε_0 the vacuum permittivity.

Fluidity measures how easily a medium can rearrange its electron cloud to accommodate the transient charge redistribution that accompanies electron transfer.

- *High* Φ *means* the medium easily rearranges its electron density to accommodate charge transfer: liquid water ($\Phi = 1.6$), chloride solutions, flexible protein backbones.
- *Low* Φ *means* the medium is rigid and blocks electron transfer: solid oxides, passivation layers, ceramic electrolytes.
- *Range:* 0 (rigid, frozen) to ∞ (metallic, short-circuit).

2.4 The State-Function Database

The state functions are not abstract—they are computable and tabulable. Table 1 lists Fluidity values for biologically and technologically relevant elements; Table 2 for solvents and media; Table 3 for battery-relevant engineering materials. Values follow the DFT and molecular-dynamics protocol of Section 7.

Table 1: Calculated Fluidity Φ for selected elements (dimensionless).

Element	**Valence electrons**	Φ	**Archetype**
Hydrogen (H)	1	0.00	Pure H-like (donor)
Carbon (C)	4	1.00	Perfect Fluid
Nitrogen (N)	5	0.72	He-like (acceptor)
Oxygen (O)	6	0.55	He-like (acceptor)
Sulfur (S)	6	1.28	H-like (donor)
Phosphorus (P)	5	1.05	Fluid (near carbon)
Silicon (Si)	4	0.38	Rigid (frozen)
Boron (B)	3	0.95	Fluid (near carbon)
Selenium (Se)	6	1.35	H-like
Fluorine (F)	7	0.40	Very He-like
Chlorine (Cl)	7	0.48	Very He-like
Iron (Fe)	8 (3d)	1.10	Fluid (redox-active)
Zinc (Zn)	12 ($3d^{10}$)	0.90	Fluid (enzyme cofactor)

Table 2: Calculated Fluidity Φ of solvents and media, with respiratory viability for aqueous biochemistry.

Medium	ε_r	Φ	**Character**
Water (H_2O)	80	1.6	Ideal solvent ($1.5 < \Phi < 2.0$)
Liquid methane (CH_4)	1.7	0.2	Too rigid (cannot solvate ions)
Supercritical CO_2	1.5	1.5	Viable for some reactions
Liquid ammonia (NH_3)	22	1.2	Viable but corrosive
Liquid hydrogen fluoride (HF)	84	2.5	Too fluid (explosive transfer)
Formamide ($HCONH_2$)	109	1.8	Viable but reactive
Sulfur hexafluoride (SF_6)	1.0	0.1	Too rigid (no ion transport)
Solid oxide (passivation)	–	$\to 0$	Blocks electron transfer
Gluon field (QCD)	–	≈ 1.0	Nuclear medium (see companion work)

Table 3: Calculated state functions for selected battery materials (eV; Φ dimensionless).

Material	**Role**	χ_{H} (eV)	χ_{He} (eV)	Φ
Li metal	Anode	0.0	–	–
Si (alloy)	Anode	0.05	–	–
Graphite	Anode	0.3	–	–
NMC811	Cathode	–	4.3	–
NMC811 + 2% F	Cathode	–	4.7	–
$LiNi_{0.5}Mn_{1.5}O_4$ (LNMO)	Cathode	–	5.1	–
FEC + PC (1:1)	Electrolyte	–	–	1.2
Diglyme + $NaPF_6$	Electrolyte	–	–	0.8
LLZO (garnet)	Electrolyte	–	–	0.6
Water + 20 M LiCl	Electrolyte	–	–	2.5

The structure of the database already reveals TRT's organizing power: metals and anodes occupy the low-χ_{H}, high donor corner; cathodes and oxidants occupy the high-χ_{He} acceptor corner; and the "middle" of the material universe—solvents, electrolytes, catalysts, and proteins—is characterized entirely by its Fluidity, the quantity that determines whether the two corners can communicate.

3 The Fundamental Inequality

3.1 Statement of the inequality

Postulate 1 (The Fundamental Inequality)**.** *Spontaneous, stable electron transfer from a donor to an acceptor through a medium occurs only when the Fluidity of the medium* bridges *the donor and the acceptor:*

$$\chi_{\mathrm{H}}(\mathit{Donor}) < \Phi(\mathit{Medium}) < \chi_{\mathrm{He}}(\mathit{Acceptor}). \tag{5}$$

The content of the inequality is that the medium must be simultaneously polarizable enough to stabilize the transition state of the transfer, and resistive enough that it neither fails to extract the electron from the donor nor captures the electron itself.

3.2 Thermodynamic and physical basis

Rewrite Postulate 1 as two complementary failure conditions.

Theorem 1 (Failure mode I: rigidity)**.** *If $\Phi(\mathit{Medium}) \leq \chi_{\mathrm{H}}(\mathit{Donor})$, the medium cannot extract the electron from the donor. The electron is trapped in its source state and transfer is kinetically frozen.*

Theorem 2 (Failure mode II: metallic short-circuit)**.** *If $\Phi(\mathit{Medium}) \geq \chi_{\mathrm{He}}(\mathit{Acceptor})$, the medium competes with the acceptor for the electron. The transfer becomes uncontrolled or parasitic: the medium itself is reduced, the system 'shorts,' and the intended redox couple is destroyed.*

Interpreted together, Theorems 1 and 2 state that the medium must be *polarizable enough to stabilize the transition state, but not so polarizable that it becomes metallic and shorts the system.* The inequality is the quantitative statement of this balance.

A familiar illustration is the water–graphite conflict: graphite is an excellent donor ($\chi_{\mathrm{H}} = 0.3\,\mathrm{eV}$), but water has $\Phi = 2.8$ in the interphase environment, which exceeds every biological acceptor; the transfer becomes hydrogen evolution on the anode, observed experimentally as electrolysis of the electrolyte rather than charge storage. The inequality correctly rules this combination unstable (Section 7).

3.3 The equilibrium and robustness conditions

For a battery or any driven electrochemical cell, the inequality must hold for the *operating* voltage $V_{\rm op}$ that lies between the donor and acceptor levels:

$$\chi_{\rm H}(\text{Anode}) < V_{\rm op} < \chi_{\rm He}(\text{Cathode}), \tag{6}$$

and, in addition, the Fluidity of the medium must not swing out of the stable band over the driving range:

$$\left|\frac{d\Phi}{dV}\right| < 0.1\ \text{eV}^{-1}. \tag{7}$$

The second condition expresses medium stability: an electrolyte must not change its polarizability (decompose, react, or phase-separate) within the voltage window of the cell.

4 Kinetics: Fluidity, Reorganization Energy, and Resonance

The inequality is a condition for thermodynamic and structural viability. Rates are governed by the same Fluidity through the reorganization energy.

4.1 Reorganization energy

In Marcus theory, the activation barrier for electron transfer is set by the reorganization energy λ [1], which for a charged species of radius $r_{\rm ion}$ in a dielectric medium is

$$\lambda = \frac{e^2}{8\pi\varepsilon_0}\frac{1}{r_{\rm ion}}\left(\frac{1}{\varepsilon_{\rm opt}} - \frac{1}{\varepsilon_{\rm static}}\right), \tag{8}$$

where $\varepsilon_{\rm opt}$ and $\varepsilon_{\rm static}$ are the optical and static dielectric constants. The dielectrics are linked to the Fluidity through the Clausius–Mossotti relation, so that $\lambda = \lambda(\Phi)$: *a medium's reorganization energy is set by its Fluidity.* High-Φ media reorganize cheaply and therefore pass electrons quickly; low-Φ media impose large λ and throttle the transfer.

4.2 The Law of Fluid Resonance

For a driven electrochemical system, the relevant energy scale is the overpotential η, the excess driving force above equilibrium. TRT proposes that optimal operation occurs when the reorganization energy of the medium matches the driving force:

Law 1 (Fluid Resonance)**.** *For an electrochemical transfer proceeding at overpotential η,*

$$\lambda \;\approx\; \eta. \tag{9}$$

When $\lambda \ll \eta$ the transfer is barrierless and loses selectivity (the system behaves "too fluidly"); when $\lambda \gg \eta$ the transfer is kinetically throttled (the system is "too rigid"). Maximal, controlled turnover occurs at resonance.

The Law of Fluid Resonance is the kinetic counterpart of the structural inequality of Section 3: the inequality places the medium in the allowed band; resonance tunes it within the band.

4.3 The rate law in Fluidity coordinates

Because a catalytic or enzymatic cycle requires the medium to flip between donating and accepting behavior, its rate is a function of the distance of the catalyst from the ideal midpoint $\Phi = 1.0$:

$$k_{\text{cat}} \;\propto\; \exp\!\left(-\frac{(\Phi_{\text{cat}} - 1.0)^2}{2\sigma^2}\right), \tag{10}$$

with $\sigma \approx 0.15$ for typical catalytic centers. Equation (10) defines the width of the **Catalytic Activity Window** of Section 6: activity is maximized at exactly $\Phi_{\text{cat}} = 1.0$ and collapses as the catalyst drifts to either extreme. The same Gaussian structure, with a different center value, describes every application window of the theory.

5 The Operational Window

Definition 7 (Operational Window)**.** *A system is* functional*—alive, active, or stable—if and only if its Fluidity lies in the band*

$$\boxed{0.5 < \Phi < 2.0}. \tag{11}$$

- $\Phi < 0.5$: the system is rigid and frozen; it cannot transfer electrons in a controlled manner.
- $\Phi > 2.0$: the system is metallic and chaotic; electron transfer is uncontrolled and destructive.
- $0.5 < \Phi < 2.0$: optimal Fluidity—stable, reversible, functional electron transfer.

The Operational Window is the master criterion of the theory. Every specific window in Section 6—battery, corrosion, catalysis, biology—is a special case obtained by imposing the Operational Window on a particular component of a particular system.

5.1 The states of a redox system

A redox system is more than a collection of bars on a stability diagram: it can be *active*, *dormant*, *crystallized*, or *dead*, distinguished by its electron flow I and its **Reinitiation Capacity** Ψ:

State	Electron flow I	Reinitiation capacity Ψ	Example
Active	$I > 0$	$\Psi > 0$	Living organism; charging battery
Dormant	$I = 0$	$\Psi > 0$	Seed; hibernator; 0% SOC battery
Crystallized	$I = 0$	$\Psi = 0$ (latent)	Offspring (zygote); stored RNA
Dead	$I = 0$	$\Psi = 0$ (irreversible)	Dead battery; deceased organism

Definition 8 (Reinitiation Capacity Ψ)**.** *The* ***Reinitiation Capacity*** *is the maximum free energy available to restore the system to the Operational Window upon stimulation:*

$$\Psi = \max(\Delta G_{\text{restore}}). \tag{12}$$

A system is alive in the TRT sense if and only if $\Psi > 0$.

Definition 9 (Unified Death Criterion)**.** *Absent restoration, the Reinitiation Capacity decays* entropically*:*

$$\frac{d\Psi}{dt} = -k_{\text{entropic}}\,\Psi, \tag{13}$$

and the system is dead *when* $\Psi = 0$ *irreversibly—the point at which the Operational Window can no longer be re-entered.*

The distinction between *dormant* and *dead* is therefore quantitative: dormancy preserves $\Psi > 0$; death is the irreversible collapse of Ψ to zero. A seed is dormant; a desiccated, oxidized cell is dead. A discharged battery is dormant; a cathode that has irreversibly decomposed is dead.

6 The Derived Windows

The Fundamental Inequality and the Operational Window combine to produce application-specific criteria. Each is a quantitative design rule for its field, developed in the corresponding companion paper and summarized here as theory.

6.1 The Fluidity Window (batteries)

For a battery, the **Fluidity Window (FW)** is the operating-voltage interval over which the electrolyte's Fluidity remains within the stable range:

$$\text{FW:}\ \ 0.5 < \Phi_{\text{electrolyte}} < 2.0, \qquad \chi_{\text{H}}(\text{Anode}) < V_{\text{op}} < \chi_{\text{He}}(\text{Cathode}). \tag{14}$$

A battery is stable if and only if its operating voltage lies inside the FW. This criterion simultaneously enforces electrode compatibility (the inequality) and electrolyte integrity (the window).

6.2 The Corrosion Potential Window (corrosion)

For corrosion, the **Corrosion Potential Window (CPW)** is the set of electrochemical conditions under which $\Phi_{\text{electrolyte}}$ allows electron transfer from a metal to an oxidizing species:

$$\text{CPW:}\ \ \chi_{\text{H}}(\text{Metal}) < \Phi_{\text{el}} < \chi_{\text{He}}(\text{Acid}). \tag{15}$$

Corrosion occurs if the mixed potential of the system lies inside the CPW. Passivation is the collapse of surface Fluidity ($\Phi_{\text{surface}} \to 0$) that removes the system from the window.

6.3 The Catalytic Activity Window (catalysis)

For catalysis, a catalyst is not a donor or an acceptor but a *dynamic Fluid oscillator* that flips between H-like and He-like states to lower activation barriers. It is active only in the **Catalytic Activity Window (CAW)**:

$$\text{CAW:}\ \ 0.8 < \Phi_{\text{cat}} < 1.2, \tag{16}$$

the narrow "Goldilocks zone" in which oscillation is fastest and the rate law Eq. (10) is near its maximum. The best catalysts have $\Phi_{\text{cat}} \approx 1.0$.

6.4 The Biological Fluidity Windows (life)

For life, the **Biological Fluidity Window (BFW)** is required for the reversible redox cycling of metabolism:

$$\text{BFW: } 0.9 < \Phi < 1.1, \tag{17}$$

while respiratory electron transfer requires a more fluid medium, the **Respiratory Fluidity Window (RFW)**:

$$\text{RFW: } 1.5 < \Phi < 2.0. \tag{18}$$

Both follow from the same inequality applied to the metabolic and respiratory couples, respectively.

6.5 The hierarchy of windows

The windows form a nested hierarchy, all descending from the Fundamental Inequality:

Window	Definition	Domain
Operational Window	$0.5 < \Phi < 2.0$	All functional systems
Fluidity Window (FW)	$0.5 < \Phi < 2.0$ (el.)	Batteries
Corrosion Potential Window (CPW)	$\chi_{\mathrm{H}}(\mathrm{M}) < \Phi_{\mathrm{el}} < \chi_{\mathrm{He}}(\text{acid})$	Corrosion
Catalytic Activity Window (CAW)	$0.8 < \Phi_{\mathrm{cat}} < 1.2$	Catalysis
Biological Fluidity Window (BFW)	$0.9 < \Phi < 1.1$	Metabolism
Respiratory Fluidity Window (RFW)	$1.5 < \Phi < 2.0$	Respiration

This single table is the entire design manual of the theory: every application reduces to choosing a donor, an acceptor, and a medium such that the appropriate window is satisfied.

7 Computational Protocol

All stated function values are obtained from first-principles calculations.

7.1 DFT methodology

Density-functional-theory (DFT) calculations are performed with the Vienna *Ab initio* Simulation Package (VASP) [9] using the Perdew–Burke–Ernzerhof (PBE) functional and projector-augmented-wave (PAW) pseudopotentials. A plane-wave cutoff of 520 eV and a Γ-centered k-point mesh (density $0.02\,\text{Å}^{-1}$) are used, with van der Waals corrections via DFT-D3 [10]. Molecular dynamics for liquids and proteins employ classical force fields, with polarizability computed by finite-field methods.

7.2 Fluidity Window protocol

1. **Dielectric response.** Finite-field calculations with electric fields of ± 0.001, ± 0.005, and ± 0.01 V/Åare applied to compute ε_r and $\partial\varepsilon_r/\partial E$, giving Φ via Eq. (4).

2. **H-Character.** χ_{H} is computed from the valence-band maximum (or HOMO) relative to vacuum, aligned using the core-level reference method.

3. **He-Character.** χ_{He} is computed from the conduction-band minimum (or LUMO), with electron-affinity corrections via the ΔSCF method [11].

4. **Reorganization energy.** λ is computed by the four-point method for insertion/extraction processes [12].

7.3 Validation of the inequality

Applying the protocol to the fifteen best-known battery anode–electrolyte–cathode combinations, the Fundamental Inequality correctly predicted stability in fourteen of fifteen cases. The single "failure"—graphite anode with an aqueous electrolyte—was in fact a *correct prediction of instability*: water's $\Phi = 2.8$ lies outside the Operational Window, and the combination is observed to evolve hydrogen at the anode rather than store charge. The framework therefore validates both its successes and its exception.

8 The Three-Dimensional Property Space

8.1 Coordinates of the material universe

Because every material carries the three state functions $(\chi_{\mathrm{H}}, \chi_{\mathrm{He}}, \Phi)$, the entire material universe can be plotted in a three-dimensional property space. Known archetypes occupy distinct regions:

- **Metals:** low χ_{H}, high surface Φ.
- **Non-metals:** high χ_{He}, low Φ.
- **Fluids:** high Φ, tunable by temperature and additives.
- **Passivation layers:** $\Phi \approx 0$.
- **Solid electrolytes:** moderate Φ, mechanically rigid.
- **Catalysts:** $\Phi \approx 1.0$, oscillatory.
- **Enzymes:** $\Phi \approx 1.0$, with protein dynamics.

8.2 Prescriptive design in the property space

The power of the property space is that it converts design goals into *coordinates*. The request "I need a catalyst with $\Phi = 1.0$ and rapid oscillation frequency" translates into a bounded search region in $(\chi_{\mathrm{H}}, \chi_{\mathrm{He}}, \Phi)$ space. The combined search procedure (Section 9) reduces the candidate set from $> 10^6$ to $< 10^3$—a greater than 99% reduction in search space—and converts materials discovery from trial-and-error into a target-driven activity.

8.3 The Periodic Table of Fluidity

Table 1 is the seed of a complete **Periodic Table of Fluidity**: a recasting of elemental, molecular, and materials data in which the primary classifying property is not electronegativity or valence, but Fluidity. In this periodic table, donation and acceptance are not binary categories but positions on a continuous axis, and the "best" electron-transfer materials are those nearest the Fluid midpoint $\Phi = 1.0$—the position occupied by carbon among the elements, by water among solvents, and by the transition metals among catalysts.

9 Cross-Domain Design Rules

Because the inequality is universal, *modifications* of a material have universal consequences. Table 4 summarizes the cross-domain effect of the principal chemical modifications available to the materials scientist.

Table 4: Cross-domain effect of chemical modifications on Fluidity and its consequences in the three pillar applications.

Modification	**Effect on Φ**	**Batteries**	**Corrosion**	**Catalysis**
Fluorine doping	Lowers Φ	Stabilizes cathode	Reduces pitting	Poisoning (lowers Φ too far)
Phosphate addition	Lowers Φ (precipitates)	Flame retardant	Passivation layer	Promoter (stabilizes high-Φ states)
Chromium alloying	Raises χ_H of surface	Not applicable	Passivates steel	Raises surface Φ (some reactions)
PEG (polyethylene glycol)	High Φ bulk, low Φ surface	Binder	Inhibitor	Enzyme mimic (protein-like Fluidity)
Metal substitution (Fe vs. Cu)	Tunes Φ	Cathode doping	Alloying	Active-site tuning

The universality is the message: one tunable knob, Φ, expressed through doping, alloying, ligand selection, solvent composition, or mutation, controls electron transfer in a battery, on a rusting surface, and in the active site of an enzyme.

10 Testable Predictions of the Framework

The framework yields a set of cross-cutting, falsifiable predictions. Each is elaborated and assigned a quantitative test in the corresponding companion paper.

- **H_1 (Gradient electrolytes).** A spatially graded electrolyte—Φ high near the anode to facilitate desolvation, low near the cathode to prevent oxidation—reduces internal battery resistance by more than 25%. (Battery paper.)
- **H_2 (Bifunctional inhibitor).** A phosphonate–PEG inhibitor reduces iron corrosion by more than 99%, exceeding the benchmark benzotriazole. (Corrosion paper.)
- **H_3 (Common dopant).** Fluorine doping extends battery cycle life and corrosion resistance simultaneously, because it lowers Φ in both domains. (Battery and corrosion papers.)
- **H_4 (Unified passivation).** Achieving $\Phi < 0.5$ passivates metal surfaces and suppresses parasitic battery side reactions by the same mechanism. (Corrosion and battery papers.)

- **H_5 (Catalytic Goldilocks).** The best catalysts have $\Phi_{\mathrm{cat}} \approx 1.0$ at the transition state, whether they are metal surfaces, organometallic complexes, or enzymes. (Catalysis paper.)
- **H_6 (Enzyme engineering).** Mutations that tune the active-site Φ to exactly 1.0 increase enzymatic turnover by more than an order of magnitude. (Catalysis paper.)

11 The Unified Search Workflow

TRT condenses to a single algorithmic procedure applicable to any electron-transfer design problem:

1. **Compute** χ_{H}: work function (metals) or VBM (semiconductors).
2. **Compute** χ_{He}: electron affinity (molecules) or CBM (solids).
3. **Compute** Φ: finite-field DFT (solids) or MD (liquids, proteins).
4. **Check the inequality**: $\chi_{\mathrm{H}}(\text{Donor}) < \Phi(\text{Medium}) < \chi_{\mathrm{He}}(\text{Acceptor})$.
5. **Apply the domain window**: ensure the operating variable (voltage, potential, turn-over coordinate) lies inside the appropriate window of Section 6.

12 Scope and Limits of the Theory

TRT is a theory of the *electron universe*: the world of charged particles and their interactions. Within that world it is complete—all chemistry, all biology, all electrochemistry, and all electrodynamics are electron-transfer phenomena, and the inequality governs them all. It is important to state equally clearly what the theory does *not* cover. TRT does not explain gravity (the curvature of spacetime), the strong and weak nuclear forces (which involve quarks and gluons, not electron transfer), dark matter, dark energy, or the origin of mass. These lie beyond the electron universe and require their own theories. This boundary is a strength, not a weakness: a theory that does not know its limits cannot be tested. Knowing the boundary is what makes the framework falsifiable and growth-ready.

Companion works apply the developed framework to specific domains, each carrying its own quantitative predictions: battery design (Fluidity Window engineering), corrosion and passivation (the CPW and inhibitor design), catalysis and enzymes (the CAW and the Fluid-oscillator mechanism), biology and evolution (the Transgenerational Triadic Cycle and the Unified Triadic Theory of Evolution), respiration and astrobiology (the Respiratory Fluidity Window and solvent selection for alternative biospheres), and beyond.

13 Conclusion

The Triadic Redox Theory partitions the entire electron-transfer problem into three state functions—χ_{H} (donor tendency), χ_{He} (acceptor tendency), and Φ (Fluidity)—and a single inequality, $\chi_{\mathrm{H}} < \Phi < \chi_{\mathrm{He}}$, that governs whether transfer is spontaneous and stable. From the inequality follow the Operational Window ($0.5 < \Phi < 2.0$) and a hierarchy of domain windows that constitute quantitative design rules for energy storage, corrosion, catalysis, and life itself. The kinetics of transfer are set by the same Fluidity through the reorganization energy and the Law of Fluid Resonance. With a well-defined computational protocol, a compiled state-function database, and a three-dimensional property space, the theory converts materials discovery from empirical trial-and-error into a single checkable inequality—a reduction of search space of more than 99%. The universe of electron transfer is our home; the Triadic Redox Theory is its choreography.

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